

Python Tuples: Methods & Operations

Tuple Methods

1. count(x):

- Description: Returns the number of occurrences of x in the tuple.

- Example:

```
t = (1, 2, 2, 3)
```

```
t.count(2) -> 2
```

2. index(x):

- Description: Returns the index of the first occurrence of x.

- Example:

```
t = (5, 3, 7, 3)
```

```
t.index(3) -> 1
```

Tuple Operations

1. Accessing Elements:

```
t = (10, 20, 30)
```

```
t[1] -> 20
```

2. Slicing:

```
t = (10, 20, 30, 40, 50)
```

```
t[1:4] -> (20, 30, 40)
```

3. Concatenation:

```
a = (1, 2)
```

```
b = (3, 4)
```

```
a + b -> (1, 2, 3, 4)
```

Python Tuples: Methods & Operations

4. Repetition:

```
t = (1, 2)
```

```
t * 3 -> (1, 2, 1, 2, 1, 2)
```

5. Membership Test:

```
t = (10, 20, 30)
```

```
20 in t -> True
```

```
50 not in t -> True
```

6. Length:

```
t = (1, 2, 3, 4)
```

```
len(t) -> 4
```

7. Iteration:

```
for item in (1, 2, 3):
```

```
    print(item)
```

8. Nested Tuples:

```
t = (1, (2, 3), (4, 5))
```

```
t[1][1] -> 3
```

Tuple <-> List Conversion

1. Tuple to List:

```
t = (1, 2, 3)
```

```
list(t) -> [1, 2, 3]
```

2. List to Tuple:

```
lst = [4, 5, 6]
```

```
tuple(lst) -> (4, 5, 6)
```